(Please write your Enrollment Number) Enrollment No. 02701022022

End-Term Examination
(CBCS)(SUBJECTIVE TYPE)(OffLine)
Course Name:<B.Tech CSE/ECE>, Semester:<II>
(July, 2023)

Subject Code: BMA 110	Subject: Engineering Mechanics

Time :3 Hours Maximum Marks :60

Note: Q. 1 is compulsory. Attempt one question each from the Units I, II, III & IV.

Q1		(10X 2=20)	

A. Write the correct option in your answer sheet. **(1 mark each)**

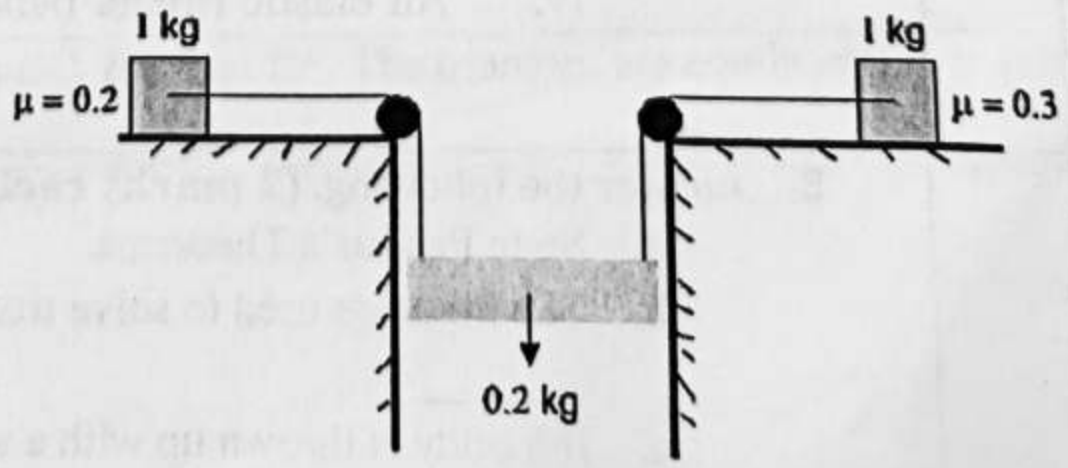
a. For analysis purposes, a body can be regarded as rigid if
   i. It has continuous distribution of mass,
   ii. It is characterized by some preferred configuration of its own
   iii. Relative motion between its parts is negligible
   iv. Its dimensions are small as compared with the co-ordinates of its own

b. A body in equilibrium is subjected to only two forces, then the only requirement is that
   i. the forces must be of equal magnitude
   ii. the forces must be equal in magnitude and opposite in direction
   iii. the forces must be collinear, equal in magnitude and opposite direction
   iv. the forces must be equal in magnitude and perpendicular to each other

c. The three coplanar and concurrent forces can never be in equilibrium if their magnitudes are
   i. 3, 4, 5 N
   ii. 4, 7, 12 N
   iii. 1, 1, 1 N
   iv. 1, 2, 3 N

d. The number of members in a simple plane truss cannot be
   i. 3
   ii. 4
   iii. 5
   iv. 13

e. In this case,
   i. both the blocks will slip
   ii. only left block will slip



iii. only right block will slip
iv. none of the blocks will slip

f. Three blocks A, B, C each of 1 kg mass are resting on a rough table. The contact area of each block is a square. The contact areas of the blocks are 1 cm2, 2 cm2, 3 cm2 respectively. The blocks are of the same material and their contact forces have same surface roughness. When the same forces are applied on the blocks, the magnitude of friction forces in descending order will be
   i. A, B, C
   ii. C, A, B
   iii. A, C, B
   iv. friction forces on all blocks will be equal

g. A particle is moving in a circular path with non-uniform speed. Then,
   i. Its velocity is necessarily tangential to the path and acceleration is normal to the path.
   ii. Its velocity is necessarily tangential to the path, but the acceleration has normal and tangential compact.
   iii. Its velocity has normal and tangential compact.
   iv. Its acceleration is tangential

h. The following statement is **incorrect** for a rigid body in plane rotational motion:
   i. All points on the body have the same angular displacement.
   ii. All points on the body have the same angular velocity.
   iii. All points on the body have the same linear velocity.
   iv. All points on the body have the same angular acceleration.

i. The angular speed of a seconds' hand of a clock is
   i. π rad/sec
   ii. $\pi/6$ rad/sec
   iii. $\pi/15$ rad/sec
   iv. $\pi/30$ rad/sec

j. Which of the following can be considered a rigid body:
   i. A 300mm diameter and 1m long axial rod of mild steel subjected to 1N force.
   ii. water
   iii. Air
   iv. An elastic rubber band

B. Answer the following. **(2 marks each)**
   a. State Pappus's Theorems.
   b. The methods used to solve truss problems are _____ and _____
   c. If a body is thrown up with a velocity of 9.8m/s, what will be the maximum height attained by it?
   d. If a car is taking a turn, how many accelerations are acting on the car? Name them.

	e. A horse pulling a cart exerts a steady horizontal pull of 300 N and walks at the rate of 4·5 km/h. How much work is done by the horse in 5 minutes?	
	UNIT-I	
Q2	The 500-kg uniform beam is subjected to the three external loads shown. Compute the reactions at the support point *O*. The *x-y* plane is vertical.	(10)

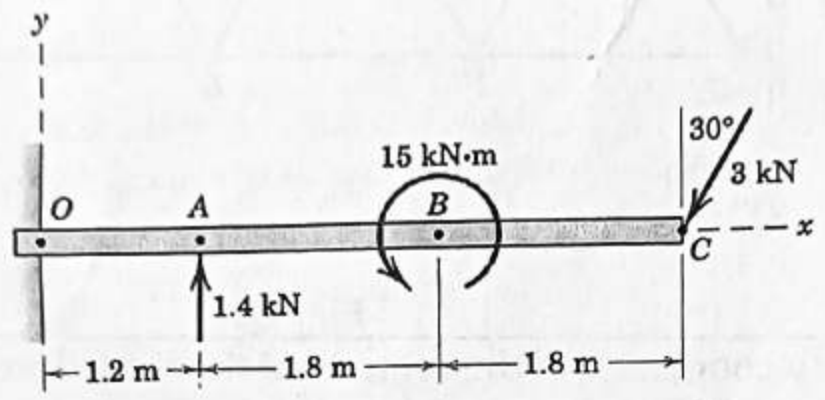


Fig: Q.2

Q3	Determine the distance *s* to which the 90-kg painter can climb without causing the 4-m ladder to slip at its lower end *A*. The top of the 15-kg ladder has a small roller, and at the ground the coefficient of static friction is 0.25. The mass center of the painter is directly above her feet.	(10)

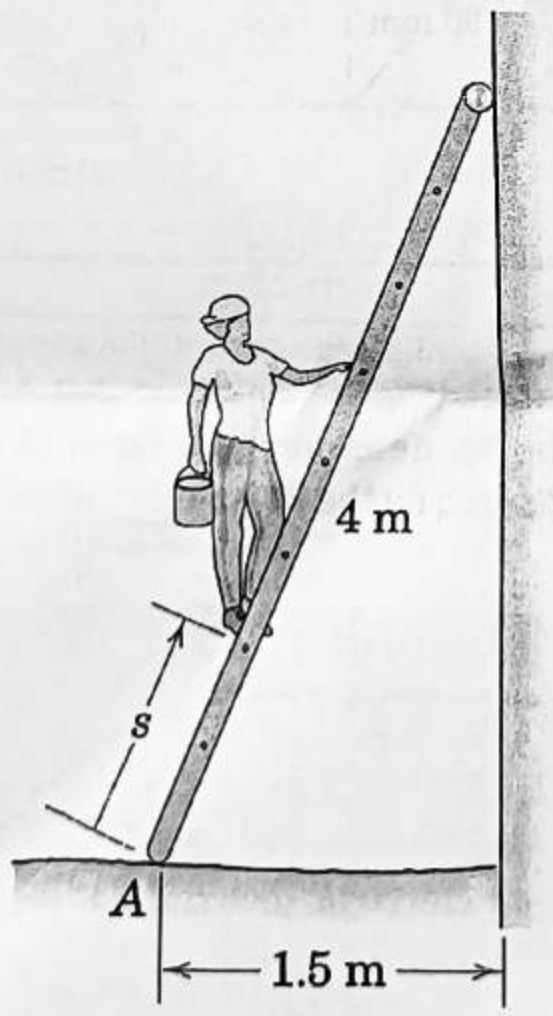


Fig: Q.3

	UNIT-II	
Q4	Determine the forces in members *BC*, *BE*, and *BF*. The triangles are equilateral.	(10)

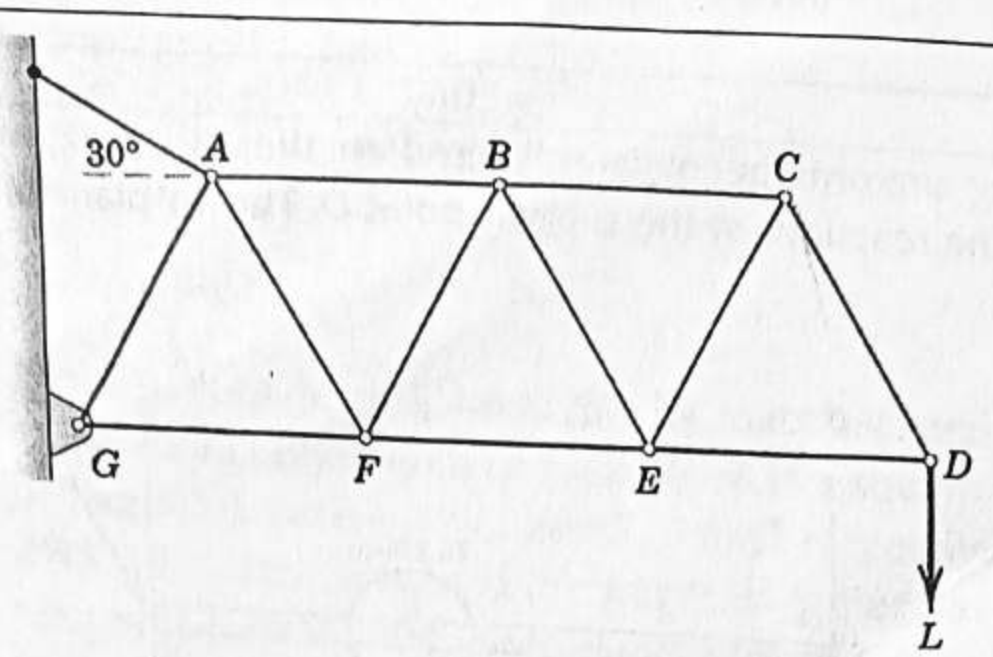


Fig: Q.4

Q5	Determine the *y*-coordinate of the centroid of the shaded area.	(10)

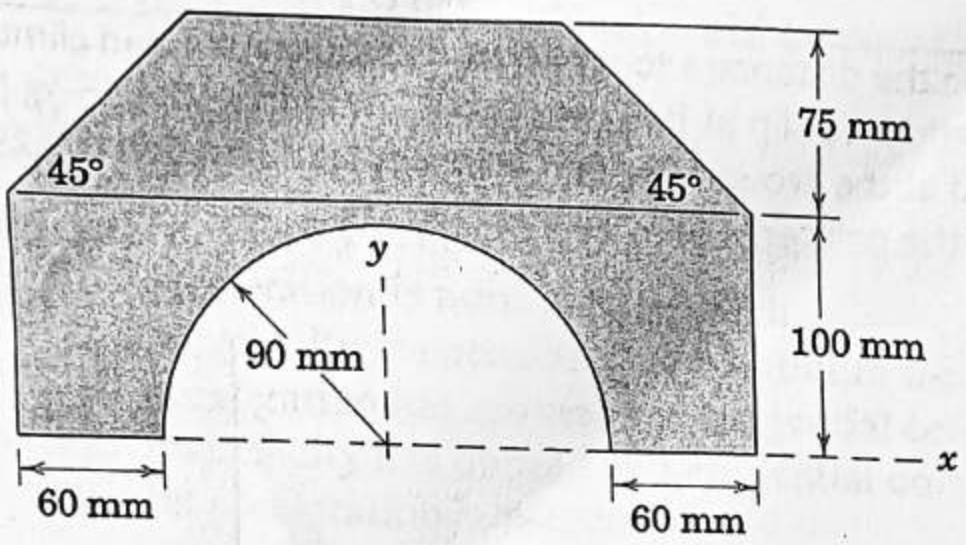


Fig: Q.5.

UNIT-III

Q6	Race car *A* follows path *a-a* while race car *B* follows path *b-b* on the unbanked track. If each car has a constant speed limited to that corresponding to a lateral (normal) acceleration of 0.8*g*, determine the times *tA* and *tB* for both cars to negotiate the turn as delimited by the line *C-C*.	(10)

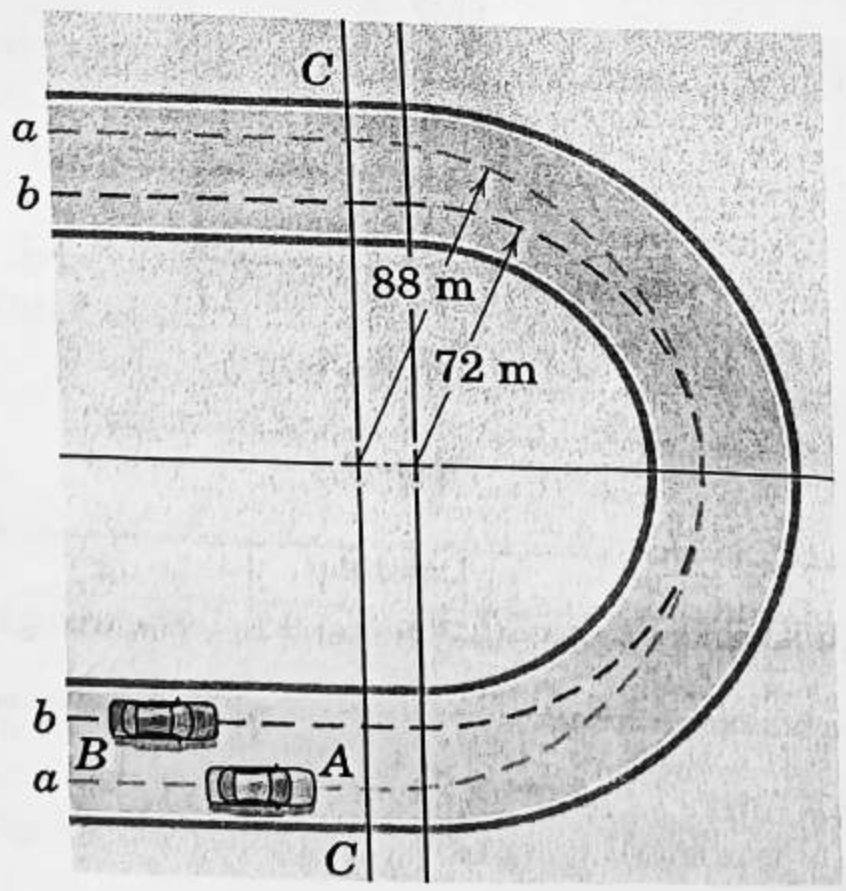


Fig: Q.6.

Q7	If the coefficients of static and kinetic friction be- tween the 20kg block *A* and the 100-kg cart *B* are both essentially the same value of 0.50, determine the acceleration of each part for (*a*) *P* =60 N and (*b*) *P* =40 N.	(10)

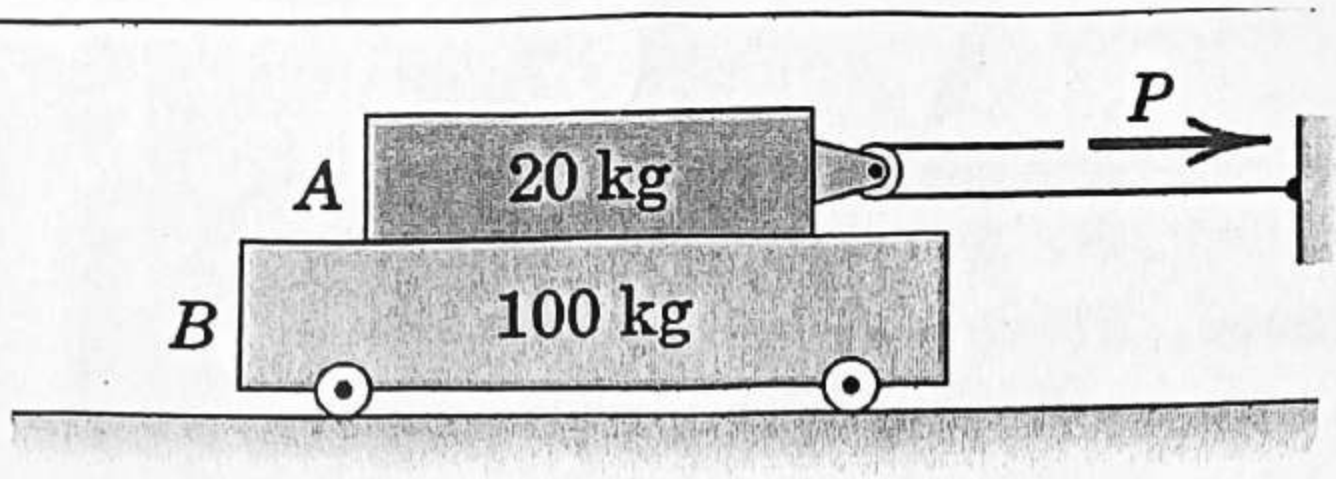


Fig: Q.7.

UNIT-IV

Q8	Determine the angular velocity of the telescoping link *AB* for the position shown where the driving links have the angular velocities indicated.	(10)

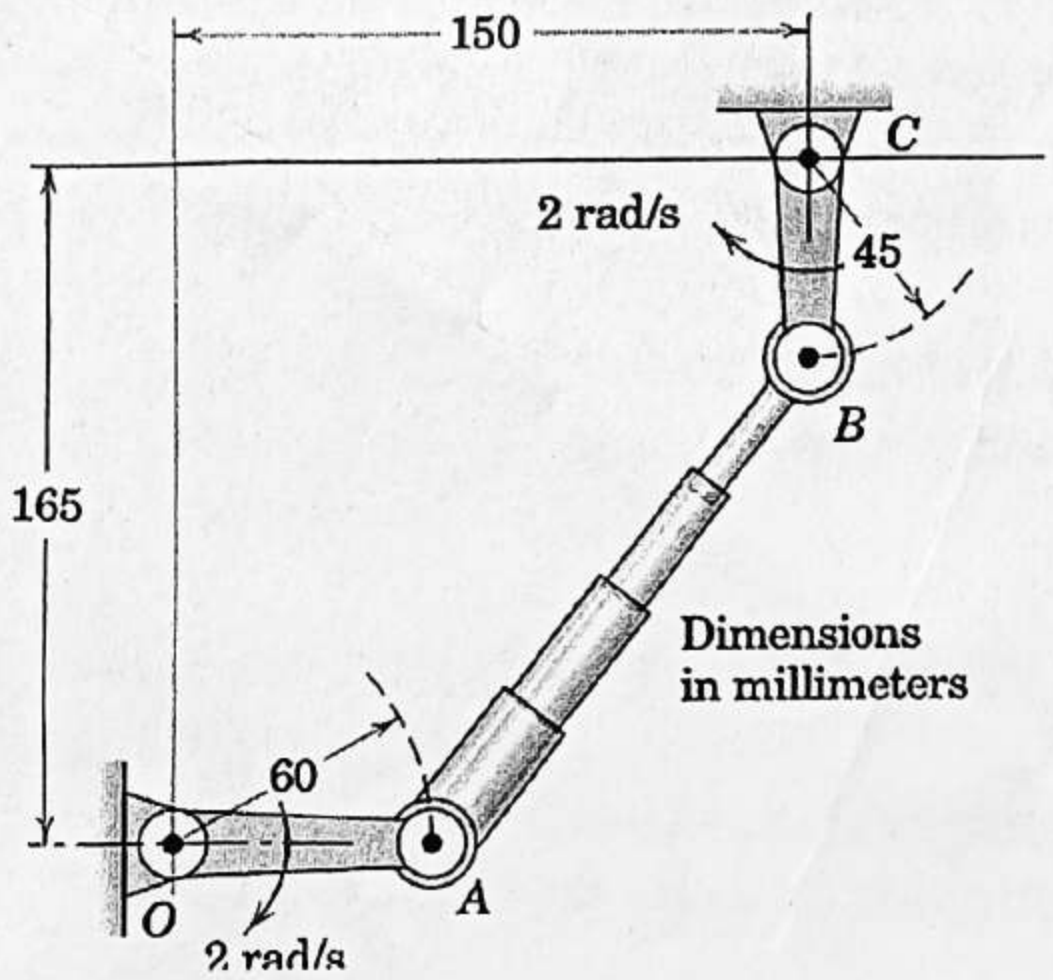


Fig. Q.8.

Q9	Horizontal motion of the piston rod of the hydraulic cylinder controls the rotation of link *OB* about *O*. For the instant represented, *vA*= 2 m/s and *OB* is horizontal. Determine the angular velocity of *OB* for this instant .	(10)

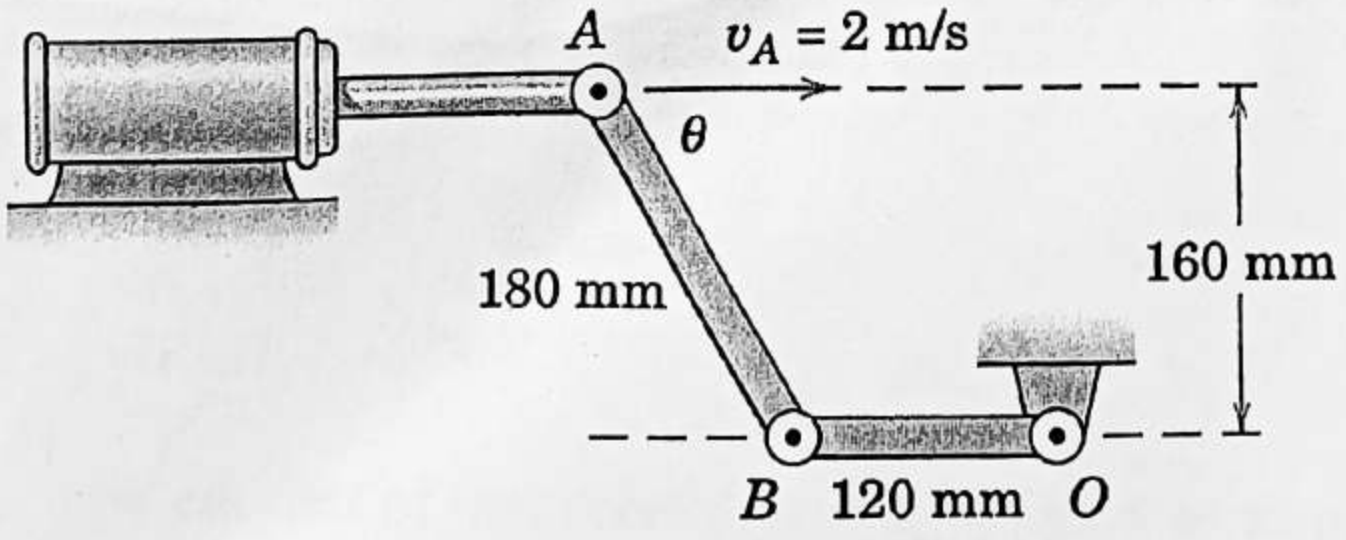


Fig. Q.9.